

Bistro Go

A Restaurant Ordering System

Project Proposal • Entity-Relationship Diagram • Project Backlog

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1. Project Proposal

1.1 Overview

Bistro Go is a web-based ordering system for a single-location, independent restaurant that currently relies on phone-in and walk-up orders. Phone orders are slow, error-prone, and interrupt kitchen staff during service. Bistro Go lets guests browse the menu and place takeout orders online, gives kitchen staff a clear queue of what to cook, and gives the restaurant owner a simple dashboard for tracking revenue, costs, and order activity — without needing a developer for day-to-day changes like updating the menu.

1.2 Users & Roles

- Guest — a customer who browses the menu and places a takeout order. No account or login is required.
- Cook — kitchen staff who need to see incoming orders in order to prepare food.
- Owner — the restaurant manager/admin who manages the menu and monitors business performance.

1.3 Goals

- Reduce order errors and phone interruptions by moving takeout ordering online.
- Give kitchen staff a live, accurate queue of orders to prepare.
- Give the owner one place to see revenue, costs, and order volume.
- Let the owner manage the menu (add items, mark items sold out) without engineering help.
- Set guest expectations clearly with order confirmation and a chosen pickup time.

1.4 Main Features

Features are grouped below by the part of the system they belong to; each maps to one or more stories in the project backlog (Section 3).

Guest ordering

- View the full menu, organized by category, with descriptions and prices (Story 1).
- Add items to an order and submit it for takeout (Story 2).
- Choose a pickup time slot at checkout (Story 5).
- See an on-screen confirmation once the order is placed (Story 9).
- Rate the order after picking it up (Story 6).

Kitchen operations

- View incoming guest orders in real time so the kitchen knows what to prepare (Story 3).

Owner tools

- Dashboard summarizing revenue, costs, and orders (Story 4).
- Mark menu items available/unavailable to hide sold-out dishes (Story 7).
- Add and edit menu items directly, with no developer involved (Story 8).
- See the current date and time on the dashboard for quick reference (Story 10).

1.5 Scope

In scope (v1): Guest menu browsing, takeout ordering, pickup time selection, order confirmation, post-pickup rating, a cook order queue, and an owner dashboard covering menu management, item availability, and revenue/cost/order summaries.

Out of scope (v1): Online payment processing, delivery/courier logistics, guest accounts or order history, loyalty/rewards programs, multi-location support, and ingredient-level inventory tracking. These are reasonable candidates for a later phase.

1.6 Proposed Technical Approach

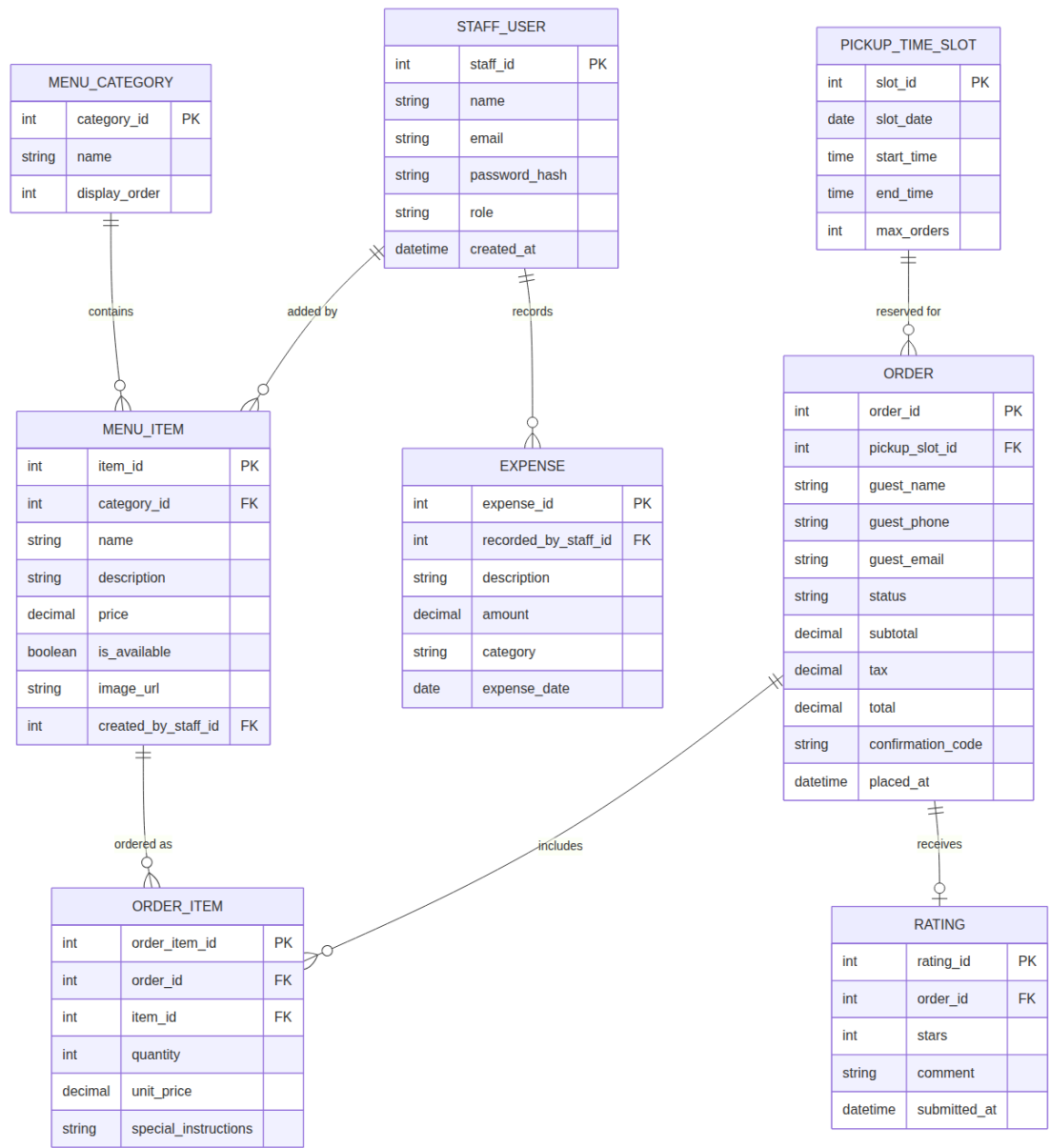
Bistro Go is a responsive web application: guests order from a phone or any browser without creating an account (contact details are captured per order to keep checkout quick), while cooks and the owner sign in to role-specific views. Application data is stored in a relational database so that menu items, orders, pickup slots, ratings, and expenses stay consistently linked — the structure of that database is shown in the Entity-Relationship Diagram in Section 2.

1.7 Success Criteria

- A guest can browse the menu, place an order, pick a pickup time, and see a confirmation without staff help.
- A cook can see a new order within seconds of it being placed.
- The owner can see today's revenue, costs, and order count from a single dashboard screen.
- The owner can add a menu item or mark one unavailable without any code changes.

2. Entity-Relationship Diagram

The diagram below shows the eight core tables behind Bistro Go and how they relate. Guests never need an account: contact details are captured directly on the Order. Revenue is calculated by summing Order totals over a date range rather than being stored on its own; day-to-day costs are captured in the Expense table — together these give the owner's dashboard (Story 4) revenue, costs, and orders in one place.



2.1 Table Definitions

Field-level detail for each table shown in the diagram above.

MENU_CATEGORY

Field	Type	Key	Description
category_id	int	PK	Unique identifier for the category.
name	string		Category name (e.g., "Appetizers", "Entrees").
display_order	int		Controls the order categories appear in on the guest menu.

MENU_ITEM

Field	Type	Key	Description
item_id	int	PK	Unique identifier for the menu item.
category_id	int	FK	References MENU_CATEGORY.category_id.
name	string		Item name shown to guests.
description	string		Short description of the dish.
price	decimal		Current menu price.
is_available	bool		Whether the item shows on the guest menu (Story 7).
image_url	string		Optional photo of the dish.
created_by_staff_id	int	FK	References STAFF_USER.staff_id — owner who added the item (Story 8).

ORDER

Field	Type	Key	Description
order_id	int	PK	Unique identifier for the order.
pickup_slot_id	int	FK	References PICKUP_TIME_SLOT.slot_id — the chosen pickup window (Story 5).
guest_name	string		Name given by the guest at checkout.
guest_phone	string		Contact phone number for the order.
guest_email	string		Contact email; no guest account is required.
status	string		placed, preparing, ready, completed, or cancelled.
subtotal	decimal		Sum of order items before tax.
tax	decimal		Tax applied to the order.
total	decimal		Final order total.
confirmation_code	string		Shown to the guest as proof the order went through (Story 9).
placed_at	datetime		When the order was submitted.

ORDER_ITEM

Field	Type	Key	Description
order_item_id	int	PK	Unique identifier for the line item.
order_id	int	FK	References ORDER.order_id.
item_id	int	FK	References MENU_ITEM.item_id.
quantity	int		Number of units of this item ordered.
unit_price	decimal		Price at the time of order, so later menu changes don't rewrite history.
special_instructions	string		Optional guest note (e.g., "no onions").

PICKUP_TIME_SLOT

Field	Type	Key	Description
slot_id	int	PK	Unique identifier for the pickup window.
slot_date	date		Calendar date of the slot.
start_time	time		Start of the pickup window.
end_time	time		End of the pickup window.
max_orders	int		Caps how many orders can be scheduled in this slot.

RATING

Field	Type	Key	Description
rating_id	int	PK	Unique identifier for the rating.
order_id	int	FK	References ORDER.order_id (one rating per order, Story 6).
stars	int		Guest rating on a 1–5 scale.
comment	string		Optional written feedback.
submitted_at	datetime		When the rating was submitted.

STAFF_USER

Field	Type	Key	Description
staff_id	int	PK	Unique identifier for the staff account.
name	string		Staff member's name.
email	string		Login email.
password_hash	string		Hashed login credential.
role	string		"cook" or "owner" — controls which views the account can access.
created_at	datetime		When the account was created.

EXPENSE

Field	Type	Key	Description
expense_id	int	PK	Unique identifier for the expense record.
recorded_by_staff_id	int	FK	References STAFF_USER.staff_id — owner who logged the cost.
description	string		What the expense was for.
amount	decimal		Cost amount, used in the owner's dashboard (Story 4).
category	string		e.g., ingredients, labor, utilities.
expense_date	date		Date the expense was incurred.

3. Project Backlog

The table below is the initial backlog for Bistro Go: ten user stories covering the guest, cook, and owner experience. Priority uses a simple MoSCoW scale (Must / Should / Could) to show a rough build order; this backlog is intentionally unrefined at this stage and is expected to be broken into smaller tasks and re-prioritized as the project moves forward.

#	Epic	User Story	Priority
1	Menu Browsing	As a guest, I want to see the full menu so that I can make my order.	Must
2	Ordering & Checkout	As a guest, I want to order food for takeout so that I can bring food home.	Must
3	Kitchen Operations	As a cook, I want to see orders from guests so that I know what to cook.	Must
4	Owner Dashboard	As an owner, I want to see restaurant data like revenue, costs, and orders so that I can track the status of the restaurant.	Must
5	Ordering & Checkout	As a guest, I want to choose a pickup time slot so that I can pick up my food when it's convenient.	Should
6	Post-Order Feedback	As a guest, I want to rate my order after pickup so that I can share my experience.	Could
7	Menu Management	As an owner, I want to set menu item availability so that I can hide sold-out dishes.	Should
8	Menu Management	As an owner, I want to add new menu items myself so that I don't need a developer to update the menu.	Should
9	Ordering & Checkout	As a guest, I want to see a confirmation message after placing my order so that I know it went through.	Must
10	Owner Dashboard	As an owner, I want to see the current date and time on the dashboard so that I can reference it easily.	Could